

H052. Compound, Concurrent, and Cascading Hazards in a Changing Climate: Identifying Drivers, Impacts, and Risk Implications

Title: Weather Jiu Jitsu: A 21st century paradigm to leverage nature's power to mitigate some Compound, Concurrent, and Cascading Weather Hazards in a Changing Climate

It is now well established that specific atmospheric circulation patterns or weather regimes can lead to multiple, nearly concurrent, clustered extreme weather events of different types—for example, floods, droughts, tropical cyclone landfalls, heat waves, and freezes—at continental or even global scales. I will review some of this evidence, and the disproportionate impact these spatio-temporally hyper-clustered events have on losses due to hazards—losses that overwhelm locally or regionally focused traditional structural and financial measures for hazard mitigation, contributing to a cascade of socio-economic impacts.

Recurrent events due to a persistent weather regime can, for instance, lead to long-duration flooding, such as the Mississippi River experiencing continuous flooding for over 100 days on two occasions in the last two decades, with lasting disruption and restructuring of supply chains.

As climate change projections portend a general increase in the frequency and severity of weather extremes, the importance of a global strategy to address this class of challenge increases. The challenge posed by such compound events at continental to global scales remains even if decarbonization of the global economy is successful. The limits of predictability of atmospheric circulation have also been established theoretically and demonstrated practically in the lack of skill of long-range weather prediction, despite significant efforts. The nonlinear dynamics of the weather–climate system induces intransitivity and chaos, which translate into weather regimes and varying predictability depending on the state of the system and the associated weather regime transition dynamics.

Weather Jiu Jitsu proposes to leverage this understanding of the system to control the weather by introducing appropriately timed and located nudges that use the power of the dynamics to amplify and steer the system away from “undesirable” weather regimes that harbor compound extremes. I will summarize the potential our group has demonstrated for such control using idealized models and deep learning emulators of the dynamics. I will also indicate possible physical mechanisms that can be used for nudging, and extend an invitation to join us in developing this new research area for the common good.